

# Shashank Siddagangaiah

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## PROFESSIONAL SUMMARY

**Data Engineer** with 9+ years of experience managing high-performing engineering teams and delivering scalable, cloud-native data solutions across insurance, healthcare, and e-commerce. Proven track record of defining 1-to-3-year technical roadmaps, aligning data architecture with business strategy, and ensuring operational excellence.

## TECHNICAL SKILLS

- **Architecture & Engineering:** Expert in designing Medallion (Bronze–Silver–Gold) data architectures, API-first microservices, and end-to-end ETL/ELT pipelines using AWS, Snowflake, Databricks, Python, and Java/Scala.
- **Real-Time & Batch Processing:** Adept at building scalable data models, real-time streaming applications (Kafka, Kinesis), and multi-TB data lakehouses that ensure 99.9% system reliability and meet strict SLAs.
- **AI & MLOps:** Hands-on experience with Agentic AI, LLMs (LangChain, Bedrock, Vertex AI), and production-grade MLOps pipelines.
- **Leadership:** Experienced in managing cross-functional agile teams (hiring, performance reviews, technical mentorship), enforcing data governance policies as code, and driving enterprise-wide digital transformations.
- **Languages & Paradigms:** Python (Pandas, PySpark, scikit-learn), SQL (PostgreSQL, PL/SQL, T-SQL), Java, Scala, R, Object-Oriented Programming (OOP), API-First Design
- **Cloud & Data Warehousing:** AWS (S3, Redshift, Glue, EMR, Lambda, Kinesis), Microsoft Azure (Synapse, Data Factory), Snowflake, Databricks, Microsoft Fabric, GCP
- **Big Data & Databases:** Apache Spark, Kafka, Hadoop, Hive, Delta Lake, Relational (MySQL, SQL Server, PostgreSQL, Teradata), NoSQL (MongoDB, DynamoDB)
- **AI, GenAI & BI:** LangChain, Hugging Face, AWS Bedrock, Vertex AI, TensorFlow, XGBoost, RAG, NLP, Tableau, Power BI, AWS QuickSight, Looker
- **DevOps & Methodologies:** Airflow, Docker, Kubernetes, Jenkins, Git/GitHub, CI/CD, MLflow, Agile, Scrum, SDLC, SLA Management, Technical Roadmapping

## PROFESSIONAL EXPERIENCE

### **PEMCO Insurance / Bourntec Solutions | Chicago, IL | Lead Data Engineer / AI Engineer | July 2025 - Present**

- Defined and led the 1-to-3-year technical roadmap for the analytics data infrastructure and AI capabilities, aligning engineering efforts with underwriting and retention business goals.
- Led the design of an Agentic AI framework to autonomously analyze Customer Lifetime Value (CLV). Designed API-first data microservices that dynamically retrieved and linked datasets, improving data accuracy by 25%.
- Orchestrated end-to-end Medallion architecture in Microsoft Fabric and Databricks. Ensured 99.9% data availability and managed strict Service Level Agreements (SLAs) for critical business data sets.
- Engineered automated ETL/ELT pipelines using Fabric Data Factory and Synapse Analytics, improving data scalability and cutting pipeline latency by 30% through parallelized orchestration.
- Embedded data quality validation, schema enforcement, lineage tracking, and policy-as-code controls directly into AI pipelines and Fabric Workspaces.
- Automated CLV visualization and reporting workflows using Tableau and Power BI, generating risk-adjusted dashboards that accelerated executive decision-making.

### **Kemper Insurance/Bourntec Solutions | Chicago | Data Engineering Manager / Lead Data Engineer | Aug 2023 – July 2025**

- Managed a team of 8 direct reports (Data Engineers and Analysts) in an Agile environment. Oversaw hiring, quarterly performance reviews, technical roadmap execution, and mentorship in cloud data engineering.
- Architected end-to-end data warehouse solutions in Snowflake and implemented real-time data streaming pipelines using Apache Kafka, Spark Streaming, and AWS Kinesis for ingestion of third-party vendor data.
- Developed scalable, object-oriented data applications and reusable pipeline components using Java, Scala, and Python, standardizing transformation logic and cutting development effort by 25%.
- Led modernization of the enterprise data ecosystem, migrating 70+ legacy MicroStrategy/OBIEE reports to Power BI and Informatica Cloud, cutting licensing costs by 40% and improving report refresh performance by 50%.
- Engineered pipeline orchestration using Airflow and custom schedulers, enhancing system reliability, reducing downtime by 30%, and defining operational SLAs.
- Instituted robust PII data masking strategies in Snowflake and Informatica, implementing role-based access control (RBAC) to align with enterprise privacy requirements (HIPAA, GDPR).

### **Select Rehabilitation | Chicago, IL | *Data Engineer / Data Analyst | May 2022 – Aug 2023***

- Designed scalable ETL/ELT pipelines using Azure Data Factory, Databricks, Spark, and AWS Glue.
- Managed unstructured and semi-structured healthcare data using MongoDB and transformed it into relational star-schemas for real-time claims and operational reporting.
- Defined and maintained strict SLAs for quarterly CMS reporting, accelerating processing cycles and improving overall data accessibility by 40%.
- Developed containerized data-processing applications using Docker and Kubernetes, improving scalability, deployment automation, and microservice orchestration.
- Migrated 100+ IBM Cognos BI reports to an in-house Python-based platform, cutting reporting costs by 20% and improving delivery speed.

### **Amazon (Pricing Analytics & Amazon Fresh Team) | Bangalore, India | *Sr Pricing Data Analyst | May 2016 – Dec 2020***

- Partnered closely with Data Scientists, Product Managers, and Software Engineers to design data architectures that supported the global expansion of Amazon Fresh (US, UK, Japan, Germany).
- Developed backend data services and automated Python-based pricing analytics pipelines, improving data refresh efficiency by 30% and enabling near-real-time business insights.
- Managed the delivery of impactful Tableau dashboards for the Weekly Business Reporting (WBR) consumed by executive leadership across 22+ business categories.

### **Amazon (Amazon Pay Team) | Bangalore, India | *Business/Data Analyst | May 2016 – Dec 2020***

- Optimized and managed massive datasets utilizing AWS Redshift and advanced SQL to support the cross-team data engineering needs of the Amazon Pay and Pricing Analytics teams.
- Led a large-scale refund-tracking data initiative during the COVID-19 crisis, consolidating fragmented unstructured data via APIs to reduce customer service contacts by over 90% (3,000 → 200 per week).

### **ACADEMIC PROJECTS**

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- **Sentiment Analysis on Yelp Reviews (NLP/R):** Predicted user ratings for new reviews using natural language processing.
- **Loan Default Prediction for Lending Club (R):** Predicted loan defaulters using Random Forest, XGBoost, and Linear Models.
- **AWS ML Deployment:** Developed and deployed a machine learning model using AWS Lambda functions, triggered via REST API.

### **CERTIFICATIONS**

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- **AWS Certified Data Engineer - Associate**
- **Tableau Desktop Specialist**

### **EDUCATION**

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**MS in Business Analytics, GPA 3.9 | *University of Illinois Chicago, Chicago, IL | Jan 2021 – May 2022***